

**Course Name: Computer Networks and Internet Protocol**

**Assignment - Week 6 (Jan 2026)**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions: 10**

**Total mark: 10 X 1 = 10**

**QUESTION 1**

In communication over an ISP, which of the following is responsible for determining the best path for packet delivery?

- a) Transport Layer
- b) Network Layer
- c) Data Link Layer
- d) Application Layer

**Correct Answer: (b)**

**Detailed solution:** The network layer, using routing protocols and IP, determines the best path for packet delivery.

**QUESTION 2**

You need 500 subnets, each with about 100 usable host addresses per subnet. What network mask will you assign using a class B network address?

- a) 255.255.255.252
- b) 255.255.255.128
- c) 255.255.255.0
- d) 255.255.254.0

**Correct Answer: (b)**

**Detailed Solution:** Class B default netmask is 255.255.0.0. To have 500 subnets, nine more bits are required to be included in the subnet mask. So, the netmask will be 255.255.255.128 (11111111 11111111 11111111 10000000). Each netmask has 7 bits for the host part, which are sufficient for 100 hosts.



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## QUESTION 3

What type of address is "0.0.0.0" in IPv4?

- a) Loopback address
- b) Default route address
- c) Broadcast address
- d) Unspecified address

**Correct Answer: (d)**

**Detailed Solution :** "0.0.0.0" is used to indicate an unspecified address, often during the bootstrapping process.

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## QUESTION 4

What is the primary function of the network layer in the TCP/IP model?

- a) Establish end-to-end connections between nodes.
- b) Ensure reliable data transfer between devices.
- c) Handle logical addressing and routing of data.
- d) Encrypt and decrypt data for secure transmission.

**Correct Answer: (c)**

**Detailed solution:** The network layer is responsible for logical addressing (e.g., IP addresses) and routing data packets across networks.

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## QUESTION 5

When communicating between two nodes, what role does the transport layer protocol play?

- a) Routing the data packets.
- b) Providing logical addressing to the communicating devices.
- c) Ensuring reliable or fast data delivery based on protocol choice.
- d) Forwarding the data to the next hop through the LAN.

**Correct Answer: (c)**

**Detailed Solution:** Transport layer protocols like TCP ensure reliable delivery, while UDP provides faster but less reliable delivery.

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### QUESTION 6

Which of the following is/are not a valid IPv6 address?

- a) AE82::1:800:23E7:F5DB
- b) FC80:2:7:1:800:23E7:A:F5DB
- c) DE62:6A42:1:5AC::800:23E7:F5DB
- d) FE80:2030:31:24

**Correct Answer: (d)**

**Detailed solution:** Option (d) is invalid because an IPv6 address must have 8 groups, but FE80:2030:31:24 has only 4 groups, making it incomplete. Whereas Option (a) is valid because the :: compresses one or more groups of consecutive zeros, and after expansion, the address still forms a proper 8-group IPv6 structure.

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### QUESTION 7

What is the maximum number of hosts under class B addresses?

- a) 254
- b) 65534
- c) 65535
- d) 65536

**Correct Answer: (b)**

**Detailed Solution:** In a Class B IP address, the first two octets are used to represent the network number, which leaves two octets or 16 bits to represent the host portion of the address. With 16 bits total, 2<sup>16</sup> combinations are possible, yielding 65,536 Class B addresses. Although two of those numbers, the lowest and highest values, are reserved for particular purposes. Therefore, each Class B address can support up to 65,534 hosts.

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### QUESTION 8

The header length of the IPv6 datagram is

- a) 10 bytes
- b) 20 bytes
- c) 30 bytes
- d) 40 bytes

**Correct Answer: (d)**

**Detailed Solution:** IPv6 has 40 bytes header length.

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### QUESTION 9

What is the CIDR notation for a supernet that aggregates 192.168.0.0/24 and 192.168.1.0/24?

- a) \23
- b) \24
- c) \25
- d) \22

**Correct Answer: (a)**

**Detailed Solution :** The supernet 192.168.0.0/23 includes both 192.168.0.0/24 and 192.168.1.0/24.

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### QUESTION 10

How does a Layer 3 switch differ from a Layer 2 switch?

- a) It uses MAC addresses for packet forwarding.
- b) It performs routing based on IP addresses.
- c) It cannot be used in VLAN environments.
- d) It only operates in the physical layer.

**Correct Answer: (b)**

**Detailed solution:** A Layer 3 switch performs routing using IP addresses, unlike a Layer 2 switch that relies on MAC addresses for forwarding.

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